



Online or in-person? Using a complex positive emotion to investigate methodology effectiveness

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Supported by BYU-Idaho's Office of Institutional Research FDSMR Grant and the Department of Psychology

INTRODUCTION

Experimental methods for manipulating and testing awe-related experiences have emphasized in-person laboratory procedures. With the ubiquitous personal use of smartphones, mobile devices, and the internet, the ability to reach broader samples of participants has never been greater. Online experimental methodologies provide a means to reach more individuals than could be reached in only in-person settings. Some research shows that while online research does allow broader reach and may cut costs, there are still some potential problems with response rates and fraudulent responses (Lefever, et al, 2007) which may impact experimental design. However, other studies suggest there is little difference between online or in-person approaches (Carini, et al., 2003; Knoepke, 2017).

PURPOSE: To compare the effectiveness of an online approach to typical in-person procedures for eliciting feelings of awe.

HYPOTHESIS

We suspect an online approach will not be as effective as an in-person approach to elicit feelings of awe.

METHODS

Participants

A sample of undergraduate students participated online (N=98) and in-person (N=123) in an experimental study designed to elicit feelings of awe. The sample was composed of 66.9% females and 83.7% White. The mean age was 21.85 years (SD=3.47).

Procedure

Participants were randomly assigned to one of four conditions: a nature condition or one of three social-elicitor conditions.

Measures

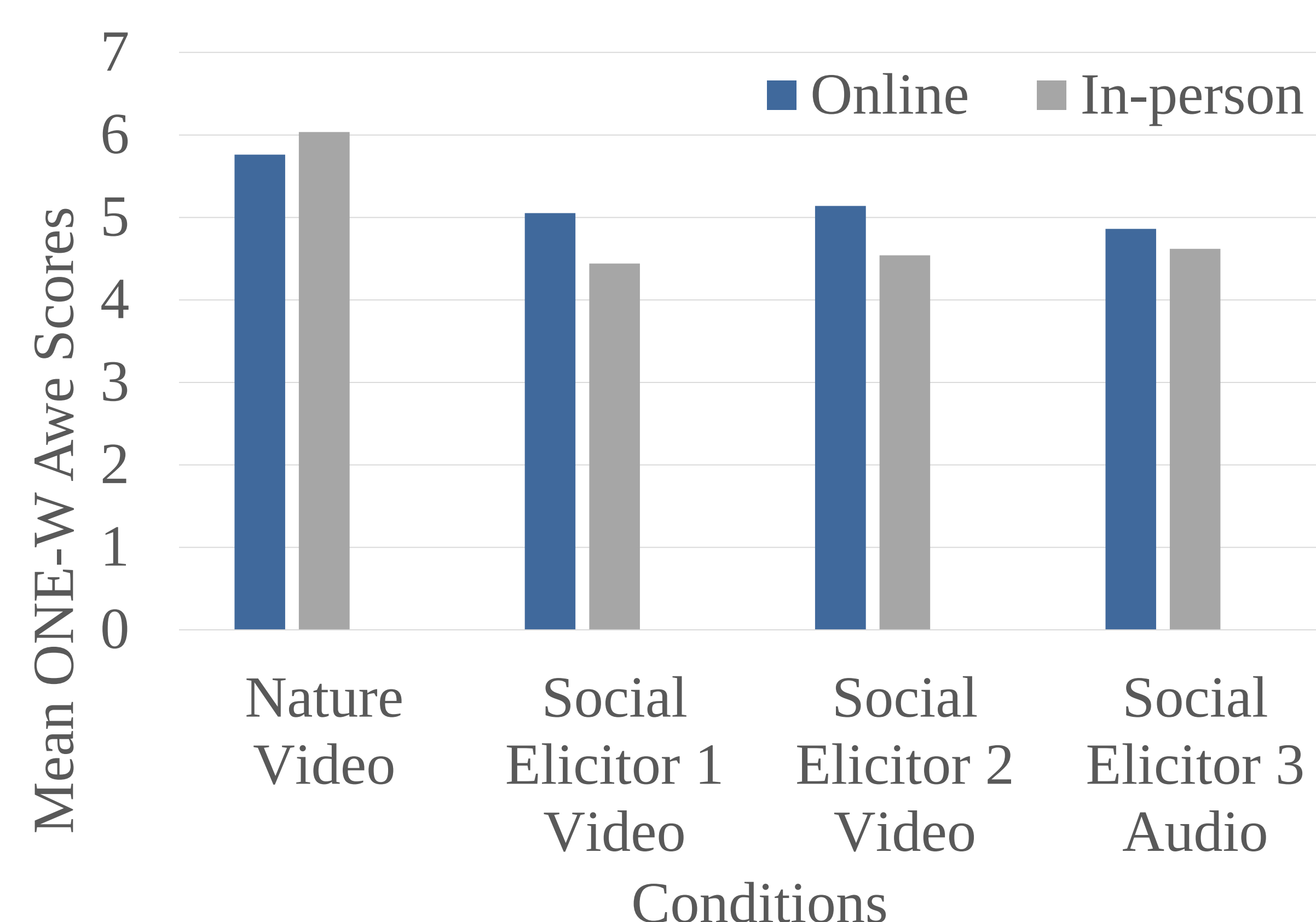
- Single Word Emotional Assessment (ONE-W):** Participants were asked to rate the appropriateness of certain words (i.e., awe) to describe their emotions during the presentation (1= not appropriate at all; to 7= very appropriate) (Shiota et al., 2007). The word “amusement” was added to the assessment.

METHODS continued

- Appraisals of Awe Related Experiences (APRS):** Participants were asked to rate their awe-related appraisals of the experience using a scale from 1 (not true at all) to 7 (very true). An example from the APRS included, “I felt small or insignificant” ($\alpha=.78$) (Shiota et al., 2007).

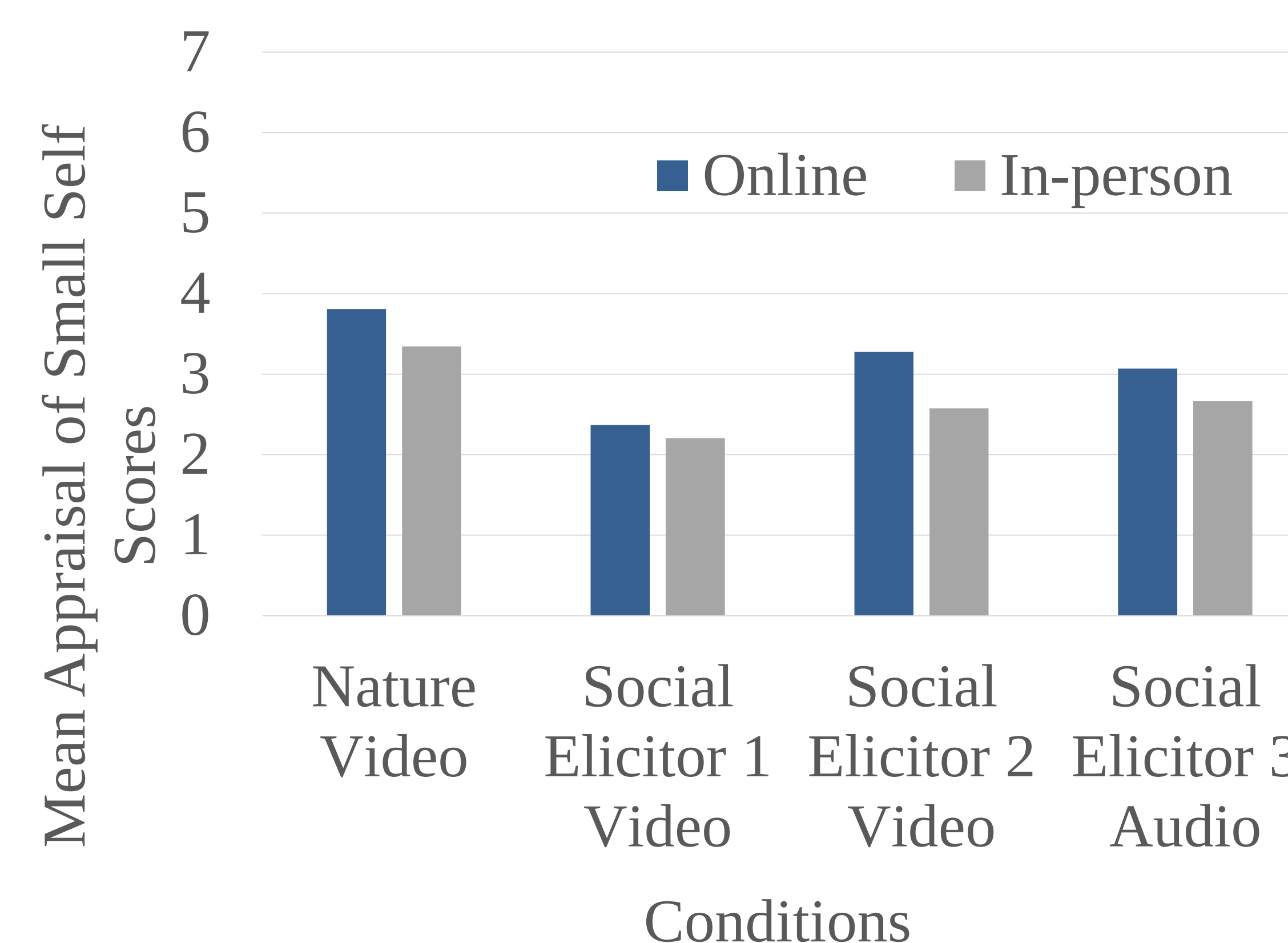
RESULTS

Figure 1. ONE-W awe scores by condition and setting.



Note: No significant differences were found between groups.

Figure 2. Appraisal of small self by condition and setting.



Note: No significant differences were found between groups.

RESULTS continued

Table 1. Comparisons between conditions for ONE-W awe

CONDITION	MAIN EFFECT/ POST HOC	P-VALUE	EFFECT SIZE
Main Effects	$F(3,213)= 6.10$	$< .001$	$\eta^2= 0.08$
Nature video – Social elicitor 1	$t= 3.51$	ns	$d= 0.76$
Nature video – Social elicitor 2	$t= 3.33$	ns	$d= 0.66$
Nature video – Social elicitor 3	$t= 3.75$	ns	$d= 0.82$
Social elicitor 1 - Social elicitor 2	$t= -0.29$	ns	$d= -0.05$
Social elicitor 1 - Social elicitor 3	$t= 0.02$	ns	$d= 0.004$
Social elicitor 2 - Social elicitor 3	$t= 0.33$	ns	$d= 0.06$

Note: Post hoc tests were done with a Scheffe correction. ns: $p > .05$

Table 2. Comparisons between online and in-person settings for ONE-W - awe

CONDITION	MAIN EFFECT/ POST HOC	P-VALUE	EFFECT SIZE
Main Effect, Setting	$F(1,213)=1.78$	ns	$\eta^2= 0.01$
In Person v Online	$t= -1.33$	ns	$d= -0.18$

Note: Post hoc tests were done with a Scheffe correction. ns: $p > .05$

CONCLUSIONS

We expected to find significant differences in study outcome scores between in-person and online study formats, but the differences were insignificant. Feelings of awe in online and in-person study methodologies did not statistically differ. Perhaps online methodologies could be more widely adopted to expand sampling pools, allow researchers to reach more diverse populations, and reduce costs. We consider results preliminary and suggest further research.